

(2) EXPONENTIAL AND LOGARITHMIC FUNCTIONS

الدوال الأسية واللوغرتمية:

With the fundamental theorems of calculus it is possible to rigorously develop the logarithmic exponential

$$(3) \quad (i) \int a^{bx} dx = \frac{a^{bx}}{b \ln a} + c \quad (a > 0, a \neq 1)$$

$$(ii) \int e^{ax} dx = \frac{1}{a} e^{ax} + c$$

$$(4) \int \ln|x| dx = x \ln x - x + c$$

Examples:

Evaluate the following integrals: (احسب التكاملات الآتية)

$$(1) \int 3^{5x} dx \quad (2) \int \left(\frac{1}{7}\right)^{5x} dx \quad (3) \int e^{8x} dx \quad (4) \int \frac{\log_3 x}{x} dx \quad (5) \int x e^{x^2} dx$$

Solution:

$$(1) \int 3^{5x} dx = \frac{e^{5x}}{5 \ln 3} + c$$

$$(2) \int \left(\frac{1}{7}\right)^{5x} dx = \int (7)^{-9x} dx = \frac{7^{-9x}}{-9 \ln 7} + c$$

$$(3) \int e^{8x} dx = \frac{1}{8} e^{8x} + c$$

$$(4) \int \frac{\log_3 x}{x} dx = \frac{1}{\ln 3} \int \frac{\ln x}{x} dx = \frac{(\ln x)^2}{2 \ln 3} + c \quad \text{where} \quad \log_3 x = \frac{\ln x}{\ln 3}$$

$$(5) \int x e^{x^2} dx = \frac{1}{2} \int x e^{x^2} d(x^2) = \frac{1}{2} e^{x^2} + c$$

Exercise (1):

Evaluate the following integrals: (احسب التكاملات الآتية)

$$(1) \int 2^{-x} dx \quad (2) \int 3x^{\sqrt{3}} dx \quad (3) \int x^{\sqrt{2}-1} dx \quad (4) \int \int_1^{\sqrt{2}} x(2)^{x^2} dx$$

$$(5) \int_0^{\frac{\pi}{2}} 7^{\cos t} dt \quad (6) \int_0^{\frac{\pi}{4}} \left(\frac{1}{3}\right)^{\tan x} dx \quad (7) \int (\sqrt{2} + 1)x^{\sqrt{2}} dx \quad (8) \int_1^e x^{(\ln 2)-1} dx$$

$$(9) \int_1^2 \frac{2^{\ln x}}{x} dx \quad (10) \int_0^3 3(1 + \ln x) dx \quad (11) \int_1^4 \frac{2^{\sqrt{x}}}{\sqrt{x}} dx \quad (12) \int \frac{\log_{10} x}{x} dx$$

$$(13) \int_1^e \frac{2 \ln 10 \log_{10} x}{x} dx \quad (14) \int_0^2 \frac{\log_2 (x+2)}{(x+2)} dx \quad (15) \int \frac{1}{x(\log_7 x)^2} dx \quad (16) \int \frac{dx}{x \log_{10} x}$$

$$(17) \int_1^{\ln 2} \frac{1}{t} dt \quad (18) \int x^5 e^{x^6} dx \quad (19) \int x(x^2 + 3)^{-4} dx$$

(3) Trigonometric functions

الدوال المثلثية:

$$(1) \int \sin x dx = -\cos x + c \quad (2) \int \cos x dx = \sin x + c$$

$$(3) \int \tan x dx = \ln|\sec x| = -\ln|\cos x| + c \quad (4) \int \cot x dx = \ln|\sin x| + c$$

$$(5) \int \sec x dx = \ln|\sec x + \tan x| + c \quad (6) \int \csc x dx = \ln|\csc x - \cot x| + c$$

$$(7) \int \sec^2 x dx = \tan x + c \quad (8) \int \csc^2 x dx = -\cot x + c$$

$$(9) \int \sec x \tan x dx = \sec x + c \quad (10) \int \csc x \cot x dx = -\csc x + c$$

Examples:

Evaluate the following integrals: (احسب التكاملات الآتية)

$$(1) \int \sin 2x dx \quad (2) \int_0^{\frac{\pi}{4}} \cos 2\pi x dx \quad (3) \int \tan x dx \quad (4) \int \cot 3x dx$$

Solutions:

$$(1) \int \sin 2x dx = -\frac{1}{2} \cos 2x + c$$

$$(2) \int_0^{\frac{\pi}{4}} \cos 2\pi x dx = \frac{1}{2\pi} [\sin 2\pi x]_0^{\frac{\pi}{4}} + c = \frac{1}{2\pi} \left(\sin \frac{\pi}{2} - \sin 0 \right) = \frac{1}{2\pi} + c$$

$$(3) \int \tan x dx = -\ln \cos x + c \quad (4) \int \cot 3x dx = \frac{1}{3} \ln \sin 3x + c$$

Example:

Evaluate the following integrals: (احسب التكاملات الآتية)

$$(1) \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{4 \cos x}{3 + 2 \sin x} dx \quad (2) \int (\tan 2x + \sec x \tan x) dx$$

Solution:

$$(1) \int_{\frac{-\pi}{2}}^{\frac{\pi}{2}} \frac{4\cos x}{3+2\sin x} dx = 2[\ln|3+2\sin x|]_{\frac{-\pi}{2}}^{\frac{\pi}{2}} = 2\ln\left|\frac{-\pi}{2}\right| - 2\ln\left|\frac{\pi}{2}\right| = 2\ln|1| - 2\ln|5| = -2\ln 5$$

$$(2) \int (\tan 2x + \sec x \tan x) dx = \int \tan 2x dx + \int (\sec x \tan x) dx = \frac{1}{2}(\ln(\sec 2x)) + \sec x + c$$

Examples:

Calculate the following integrals: (احسب التكاملات الآتية)

$$(1) \int \sec 2x dx$$

$$(2) \int \csc^2 7x dx$$

$$(3) \int \csc 9x dx$$

Solution:

$$(1) \int \sec 2x dx = \frac{1}{2} \ln|\sec 2x + \tan 2x| + c$$

$$(2) \int \csc^2 7x dx = -\frac{1}{7} \cot 7x + c$$

$$(3) \int \csc 9x dx = \frac{1}{9} \ln|\csc 9x - \cot 9x| + c$$

Example:

Prove the following integrals: (احسب التكاملات الآتية)

$$(1) \int \tan x dx = -\ln \cos x$$

$$(2) \int \sec x dx = \ln|\sec x + \tan x|$$

Solution:

$$(1) \int \tan x dx = \int \frac{\sin x}{\cos x} dx = -\int \frac{-\sin x}{\cos x} dx = -\ln \cos x$$

$$(2) \int \sec x dx = \int \frac{\sec x (\sec x + \tan x)}{\sec x + \tan x} dx = \ln|\sec x + \tan x|$$

Exercise (2):

(A) Find the simplest ant derivatives you can for each part and add the arbitrary constant of integration at the end.

احسب ابسط صورة ممكنة للتكاملات الآتية ثم اضع ثابت التكامل لكل حالة.

$$(1) \int (3x^2 + \sec x) dx$$

$$(2) \int (2x - \cos 7x + \sqrt{2x}) dx$$

$$(3) \int (\tan 6x + \sec^2 x) dx$$

$$(4) \int (5^{3x} + e^{-x} + x^{-1}) dx$$

$$(5) \int (4 - 8x)^9 dx$$

$$(6) \int \left(\frac{t\sqrt{t} + \sqrt{t}}{t^2} \right) dt$$

$$(7) \int \frac{\sec^2 x}{4} dx$$

$$(8) \int 5 \sec x \tan x dx$$

$$(9) \int 6 \cos 3x dx$$

$$(10) \int \left(x - \frac{1}{x}\right) \left(x + \frac{1}{x}\right) dx$$

$$(11) \int x^{-3}(x+1) dx$$

$$(12) \int \frac{1}{2} \sqrt[5]{x} dx$$

$$(13) \int (x-4)(x+2) dx$$

$$(14) \int \sec^2 x \tan x dx$$

$$(15) \int x^{-1} \ln x dx$$

$$(16) \int x \sinh(x^2) dx$$

$$(17) \int x \tanh(x^2 + 1) dx$$

$$(18) \int \sqrt{1 - \cos^2 x} dx$$

$$(19) \int \tan^2 x dx$$

$$(20) \int \sin^3 x \cos x dx$$

$$(21) \int \frac{e^x - 1}{e^x} dx$$

$$(22) \int x e^{x^2} dx$$

$$(23) \int e^x \cos x dx$$

$$(24) \int \cos x \sin x dx$$

$$(25) \int \frac{x^2}{x^2 + 4} dx$$

$$(26) \int \sin^3 x \cos^5 x dx$$

$$(27) \int \frac{(\sin x)^4}{\sqrt{1 - \cos^2 x}} dx$$

$$(28) \int e^x \sin(e^x) dx$$

$$(29) \int \frac{1}{x-4} dx$$

$$(30) \int \ln x dx$$

(B) **Prove that:**

$$(1) \int \cot x dx = \ln |\sin x| \quad (2) \int \tan x dx = \ln |\sec x|$$

$$(3) \int \csc x dx = \ln |\csc x - \cot x|$$

(4) HYPERBOLIC FUNCTIONS:

الدوال الزائدية